

What Survives When Activation Steering Is Used as Memory: Leanings Partly, Facts Not, and Dislikes Turn into Likes

Vedant Jumle

Independent researcher

vedantjumle@gmail.com

Abstract

1 Introduction

2 Method

3 Why a stored shift keeps some content and not other

4 Experimental setup

5 Results

5.1 The memory is selective

5.2 Facts are stored as a late word and are not used

5.3 Leanings transfer only in a narrow dose window

5.4 What the shift subtracts decides concept vs direction

6 Related work

7 Limitations

8 Conclusion

References

TODO Braun. 2025. Understanding unreliability of steering vectors in language models. Master’s thesis, University of Tübingen. TODO verify author name and form.

Yuanpu Cao, Tianrong Zhang, Bochuan Cao, Ziyi Yin, Lu Lin, Fenglong Ma, and Jinghui Chen. 2024. Personalized steering of large language models: Versatile steering vectors through bi-directional preference optimization. *arXiv preprint arXiv:2406.00045*.

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Jerry Zhi-Yang He, Sashrika Pandey, Mariah L. Schrum, and Anca Dragan. 2025. Context steering: Controllable personalization at inference time. In *The Thirteenth International Conference on Learning Representations*.

Joel Jang, Seonghyeon Ye, and Minjoon Seo. 2023. Can large language models truly understand prompts? a case study with negated prompts. In *Proceedings of the 1st Transfer Learning for Natural Language Processing Workshop*, PMLR. TODO verify volume/pages.

Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehl, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. 2025. Programming refusal with conditional activation steering. In *The Thirteenth International Conference on Learning Representations*.

Jack Lindsey. 2026. Emergent introspective awareness in large language models. Anthropic. TODO verify venue/URL.

Sheng Liu, Haotian Ye, Lei Xing, and James Zou. 2024. In-context vectors: Making in context learning more effective and controllable through latent space steering. *arXiv preprint arXiv:2311.06668*.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *First Conference on Language Modeling*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*.

Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. 2023. Mass-editing memory in a transformer. In *The Eleventh International Conference on Learning Representations*.

nostalgebraist. 2020. interpreting GPT: the logit lens. LessWrong blog post.

Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the*

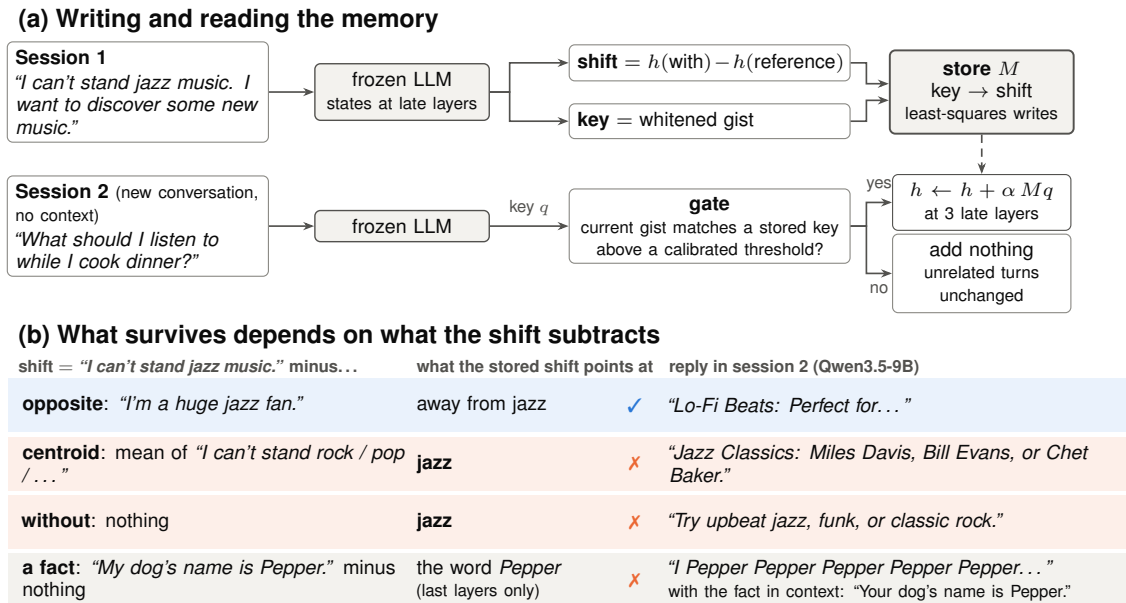


Figure 1: **(a)** A training-free activation memory beside a frozen LLM. In session 1 the change an experience causes in late-layer states is stored as a *shift*, filed under a whitened *key* for the conversation’s gist. In a later session the shift is added back only where the current gist matches a stored key. **(b)** What the stored shift carries depends on the reference subtracted when writing it. Subtracting the opposite keeps the *direction*; subtracting alternatives or nothing keeps the *concept*, so a dislike is recalled as a like. A fact is stored as the word to say. Replies are single real samples from Qwen3.5-9B (preferences $\alpha=1$, fact $\alpha=2$); quantitative results are in §5.

Association for Computational Linguistics (Volume 1: Long Papers), pages 15504–15522.

Melanie Subbiah, TODO Hall, and Kathleen McKeown. 2026. On the limits of steering vectors for preference-aligned generation. TODO verify authors and venue.

Manan Suri, Nishit Anand, and Amisha Bhaskar. 2025. Mitigating memorization in LLMs using activation steering. *arXiv preprint arXiv:2503.06040*.

Daniel Tan, David Chanin, et al. 2024. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems*. TODO verify full author list.

Chengfeng Zhao, Yuqiao Tan, Shizhu He, Yequan Wang, Jun Zhao, and Kang Liu. 2026. Neural procedural memory: Empowering LLM agents with implicit activation steering. *arXiv preprint arXiv:2606.29824*.

A Items, prompts and judge rubrics

B Judge panel and sanity check

C Additional figures

D Pre-registration and reproduction